Bt402
Microbial Biotechnology
Current Final Term Solved Spring 2022
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BT402 Current Final Term Spring 2022

Proteomics and

Proteomic analysis requires first the separation of thousands of proteins or their fragments and then their identification. It is impossible to identify this many proteins and their fragments by using the traditional approaches, such as reactivity with specific antibodies or functional properties. Thus, the advances in mass spectrometry, now almost the universal approach in identification, were the major factor that made proteomics possible. Previously, mass spectrometry, which measures the mass-to-charge ratio of ions in a vacuum required the “hard” ionization methods that fragmented the sample molecule

Transcriptomics

The presence of a gene in a genome does not obviously mean that it is expressed to produce mRNA and subsequently a protein. Furthermore, regulation of metabolism often involves alteration of expression of various genes, and therefore it is important to know their expression level in order to understand the physiology of the cell.

Before the genome sequences became available, gene expression had to be examined on a one-by-one basis. The techniques used included Northern blot, which involves annealing of fractionated mRNA with the cognate pieces of DNA, and reverse transcriptase– PCR, in which mRNA is used as a template to produce cDNA, which is then amplified by PCR

Dna vaccine

Dna vaccine

A DNA vaccine is a type of vaccine that transfects a specific antigen-coding DNA sequence into the cells of an organism as a mechanism to induce an immune response.[

• Still in the experimental stages, these vaccines show great promise, and several types are being tested in humans

. • • DNA vaccines take immunization to a new technological level

. • • These vaccines dispense with both the whole organism and its parts and get right down to the essentials: the microbe’s genetic material.

• • **Example**: Influenza vaccine.

advantages of microbial biotechnology

Microbial biotechnology is a well-established branch of biotechnology, which include the application of microorganisms with emerging modern techniques of biotechnology for the development of sustainable agriculture. Microbial biotechnology deals with the manipulation through genetic engineering of living organisms or their components to produce valuable products for various applications. Classical agriculture farming equipment and practices are reaching their limits of effectiveness in increasing agricultural productivity. The chemical fertilizers, pesticides, herbicides, and other inputs have increased agricultural production, but simultaneously, cause adverse effects on soil productivity and environmental quality. The broad application of microbes in sustainable agriculture is due to the genetic dependency of plants on the beneficial functions provided by symbiotic cohabitants. Therefore, microbial biotechnology and its applications in development of sustainable agriculture and environmental health are attracting the attention of microbiologists and biotechnologists. India's large population is placing more pressure on natural resources like soil and water, and affecting our ability to produce sufficient food, feed, and fiber. Recently developed biotechnologies can provide appropriate new tools to find out the solutions of different specific problems of sustainable agriculture (Singh and Gupta, 2018). The aim of this study was to conclude the diversified useful applications of microbial biotechnology for the development of sustainable agriculture using different tools and techniques such as biofertilizers, biopesticides, and value additions in crops

•Live-Attenuated Vaccines.

• **Live, attenuated vaccine**s contain a version of the living microbe that has been weakened in the lab so it can't cause disease.
• Because a live, attenuated vaccine is the closest thing to a natural infection, these vaccines are good "teachers" of the immune system.
• **Example**: Vaccines against measles, mumps, and chickenpox

•Inactivated Vaccines.

• Scientists produce inactivated vaccines by killing the disease-causing microbe with chemicals, heat, or radiation. Such vaccines are more stable and safer than live vaccines.
• Because dead microbes can't mutate back to their disease-causing state
. • **Example**: Vaccines against influenza, polio, hepatitis A, and rabies

Biofuels

• A biofuel is defined as any fuel whose energy is obtained through a process of biological carbon fixation.
• Biofuels are fuels produced directly or indirectly from organic material biomass , including plant materials and animal waste.

Biochemical pesticides

• They are naturally occurring substance to control pest by non-toxic mechanisms.
• Biochemical pesticides include substances as insect sex pheromones, that interfere with mating that attract insect pest to traps

What is compost

• Compost is organic matter that has been decomposed and recycled as a fertilizer and soil amendment
• Compost is a key ingredient in organic farming.
• End product of the decomposition of organic matter

Benefits of Microbes in agriculture

• Microorganisms in Agriculture
• Microorganisms have been ruling in agriculture from the past century. They are used as Biofertilizers and Biopesticides. E.g. Rhizobium, Bacillus, Azotobacter.

Unsolved Current Papers

BT402 on 19-9-22 at 11:00 am
Long:
-Main objectives of transgenic plants.
-Important role of Biofertilizers
Short:
-Challenges in Adjuvants uses.
-DNA Vaccine.
-Biofuels and their importance.
-Importance of microorganisms in plants.
-Synbiotic
-Biosurfactants
MCQs mostly ppts aur files men sy tha.
BT402 Final-Term paper 📜
18 September 2022
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🌷what is live attenuated vaccine give one example
🌷explain subunit vaccine
🌷define toxoid vaccine
🌷how microbes helpful in plant growth
🌷two types of biofertilizers
🌷 prokaryotes genome detail
🌷what is third generation biofule breif
🌷viruses genome(long)
🌷2 types of prebiotics
🌷 write ✍️ three methods of applying biofertilizers
🌷what is basic technique of genome sequencing

Ans: The techniques used included Northern blot,
which involves annealing of fractionated mRNA with the cognate pieces of DNA, and reverse transcriptase– PCR, in which mRNA is used as a template to produce cDNA, which is then amplified by PCR.

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My today paper
Bt402
McQ kuch past paper se Jo question yad kiye the UN me se the aur kuch pptz se
Short question
DNA vaccine
Live attenuated vaccine
Mechanisms of regularly process
Prokaryotes genome
Metabolism
How microbes are beneficial for biofertilizer
Second generation biofuel
What is biofuel
Long:
Inactivated vaccine
Microbial pesticide
Symbiotic, the digestion of enzyme.

bt402
Total questions 53, 40 MCQs n 13 questions.
MCQs almost 25 or 28 from mid portion n all from past files
Remaining MCQs were tough n mostly from last 20 lectures...

✅ Prokaryotic genome (2)
✅ Importance of microbes in medicine (2)
✅ Types of prebiotic (2)
✅ Y bread rise (2)
✅ Biofuel (3)
✅ DNA vaccine with ex (3)
✅d/ f b/w biomining and bio leaching (3)
✅3rd generation biofuel (3)
✅Plant-microbe interaction(3)
✅ Probiotics (3)
✅Define antimicrobial (3)

Long questions
✅ biodiesel (5)
✅dose prebiotic have side effects (5)
✅ advantage of bio pesticide (5).

Bt402. 18-sep-22 11:00 am
1. DNA vaccine with example
2. Soil treatment
3. Difference between bioleaching and biomining

4. Why it is necessary to
understand molecular mechanism of regulatory process?
5. Bio processing of microbes
6. Difference between submerged and solid state fermentation.
7.How second generation biofuels are produced?
8. Uses of antimicrobial products.
9. How yeast causes the bread to rise?
10. "Can't remember.."
11. What is biodiesel and how it is produced. 5 marks
12. Two types of bio fertilizers. 5 marks.
13. Describe the aspects of polysaccharide structure. 5 mar
Bt402 Today at 11:00am

•What are DNA Vaccines.
•Live-Attenuated Vaccines.
•Inactivated Vaccines.
•Biofuels.
•Biochemical Pesticides.
•Benifits of Microbes in agriculture.
•Role of Biotechnology in agriculture.
•Name of Fungi Involved in food/Brewing industry.
•What are Metalobomics(Aese hi kch tha).
•Benifits of Probiotics.

Upto 30 MCQS Past Se the •What is necessary to understand the molecular mechanism of regulatory process.
bT402 Final-Term 17 sep
🌷 proemics and transcriptomics

🌷Dna vaccine

🌷microbial biotechnology in health issues

🌷compost
🌷advantages of microbial biotechnology

🌷Bioleaching and microbial mining difference
🌷biofertilizer sa tha ik
🌷biopestide